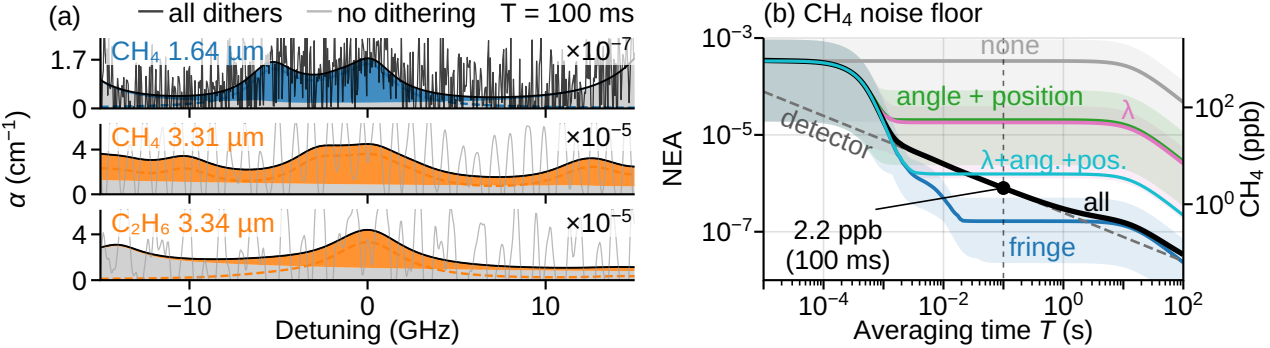

(a)
all dithers
no dithering
T = 100 ms
CH4 1.64 μm
×10−7
1.7
0
CH4 3.31 μm
×10−5
4
0
C2H6 3.34 μm
×10−5
4
0
α (cm−1)
−10
0
10
Detuning (GHz)
(b) CH4 noise floor
10−3
10−5
10−7
NEA
none
angle + position
λ
detector
λ+ang.+pos.
all
fringe
2.2 ppb
(100 ms)
10−4
10−2
100
102
Averaging time T (s)
102
100
CH4 (ppb)